

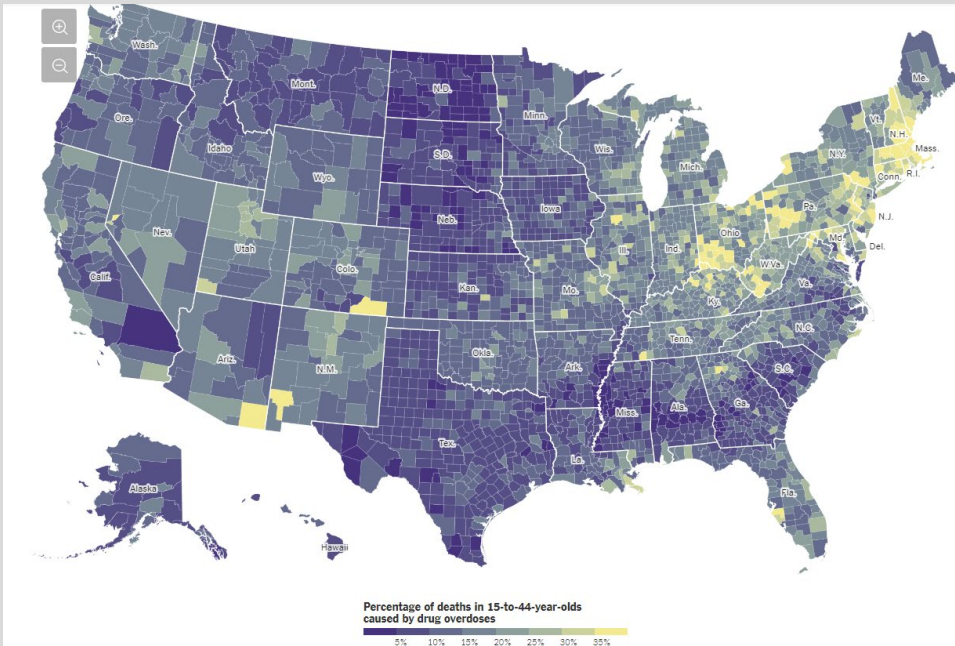
- Physiology: Drug Addiction
- **Gonzalo Otazu, Ph.D.**
- Assoc. Prof, Biomedical Sciences

Do.
Make.
Heal.

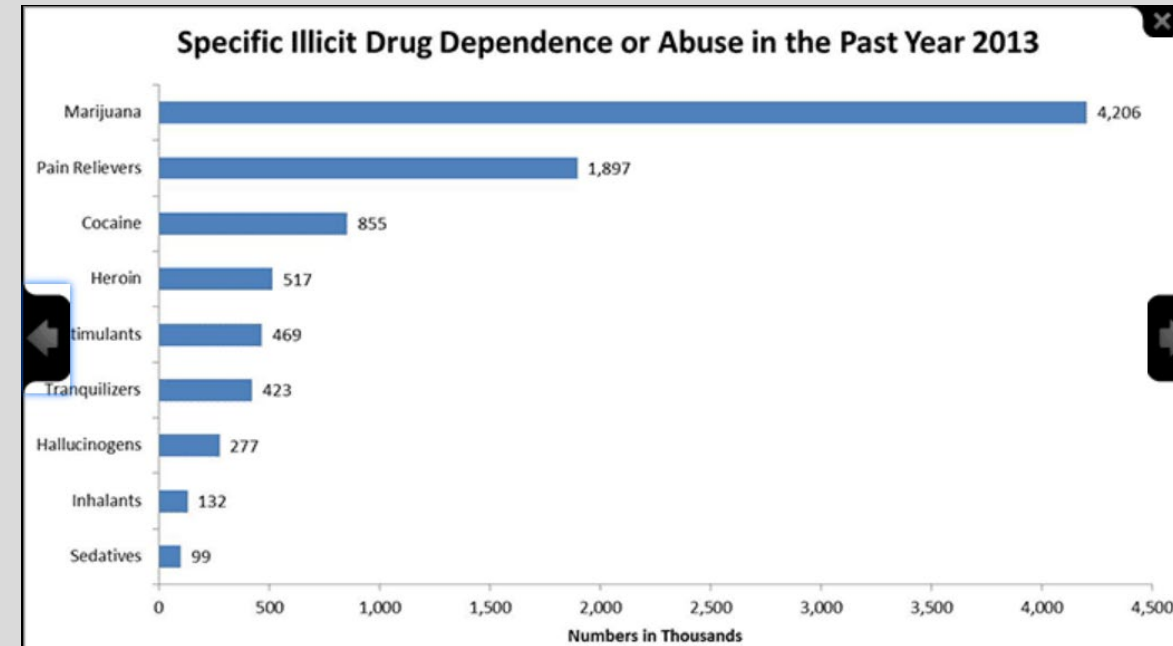
Innovate.

Reinvent the Future.

- Epidemiology of Drug Addiction
- In 2016, roughly 64,000 people died from drug overdoses; one of leading causes of **accidental death**
- Most people use drugs for the first time when they are **teenagers**
- More than half of new illicit drug users begin with **marijuana**



Cost: \$740 billion annually related to crime, lost work productivity, and health care (National Institute on Drug Abuse, 2018)



- National Institute of Drug Abuse (NIDA)

- Definitions

- **Addiction:** Addiction is a disease in which, after a period of recreational use, a subset of individuals develops compulsive use that does not stop even in light of major negative consequences. (Nestler and Luscher)

- **Reward Pathway**

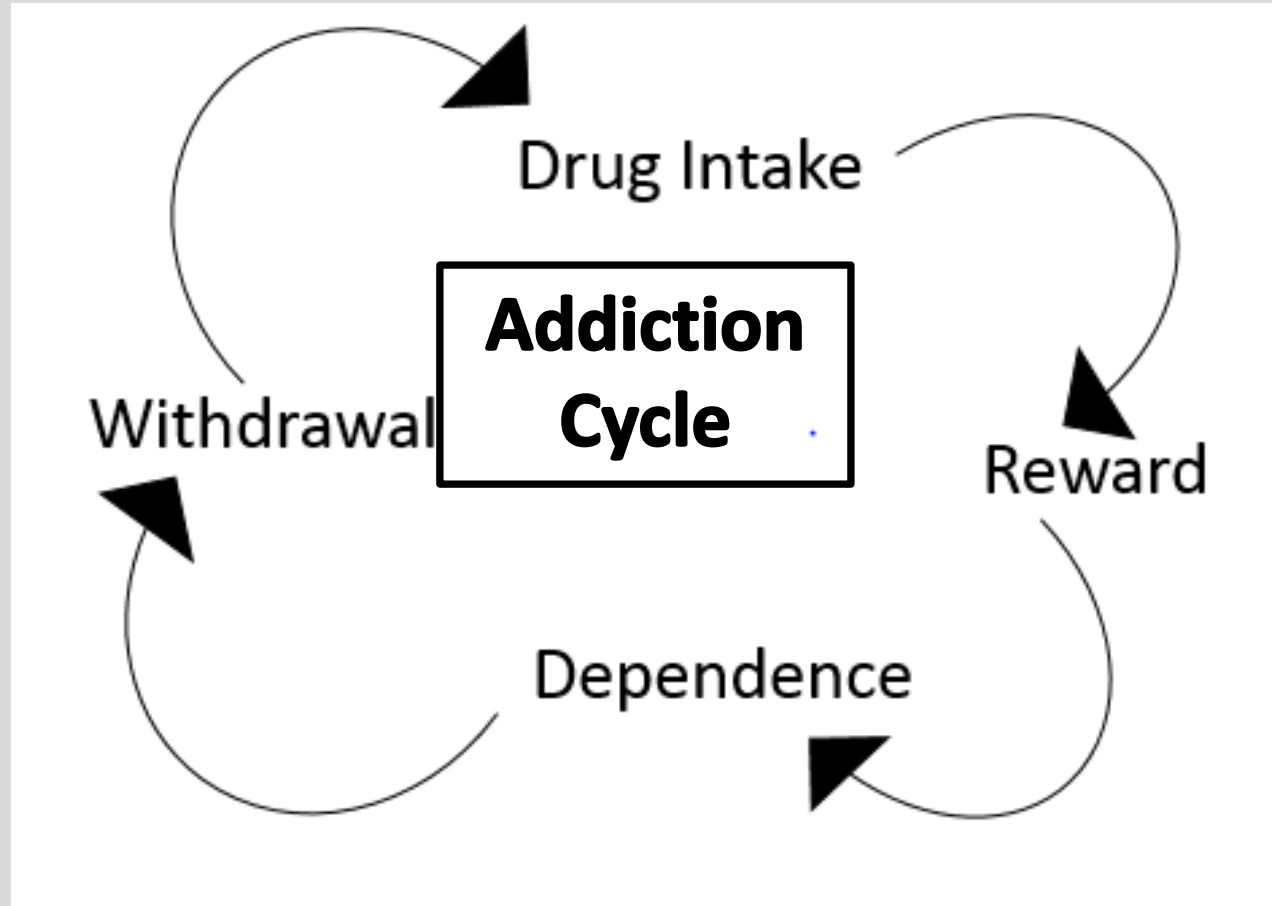
- Positive reinforcement

- **Dependence**

- Psychological
- Physical
- Craving

- **Withdrawal**

- negative symptoms associated with unfulfilled craving



- Rang & Dale's Pharmacology 8th edition

- Session Objectives

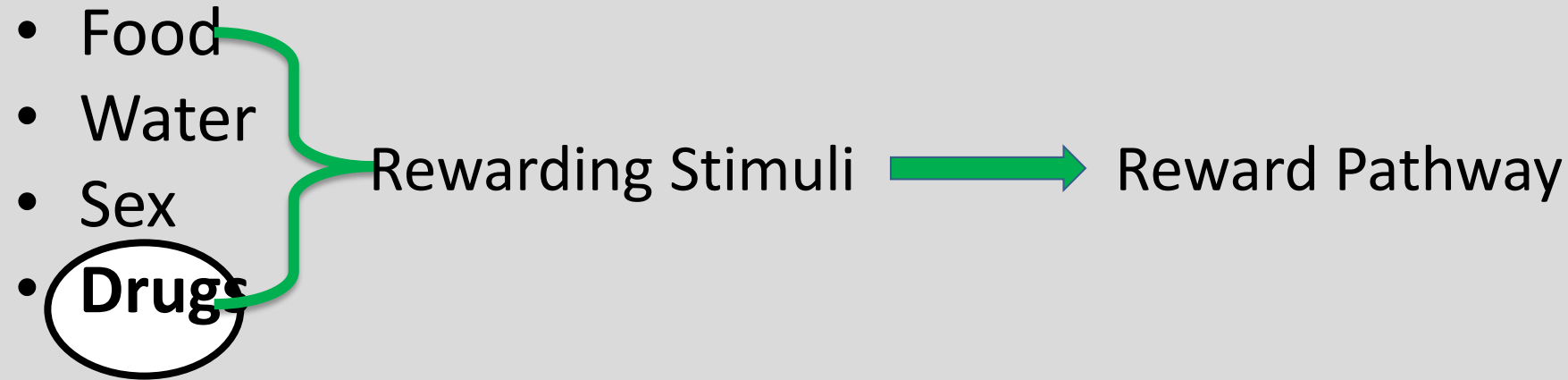
1. Students will understand the neurobiological basis of the brain's reward pathway
2. Students will become familiar with common classes of drugs of abuse (stimulants, depressants, hallucinogens.)
3. Students will understand how drug-specific pharmacological mechanisms lead to drug abuse
4. Students will be able to identify both general class and drug-specific symptoms of intoxication and withdrawal.

Do.
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- Reward Pathway: Concepts

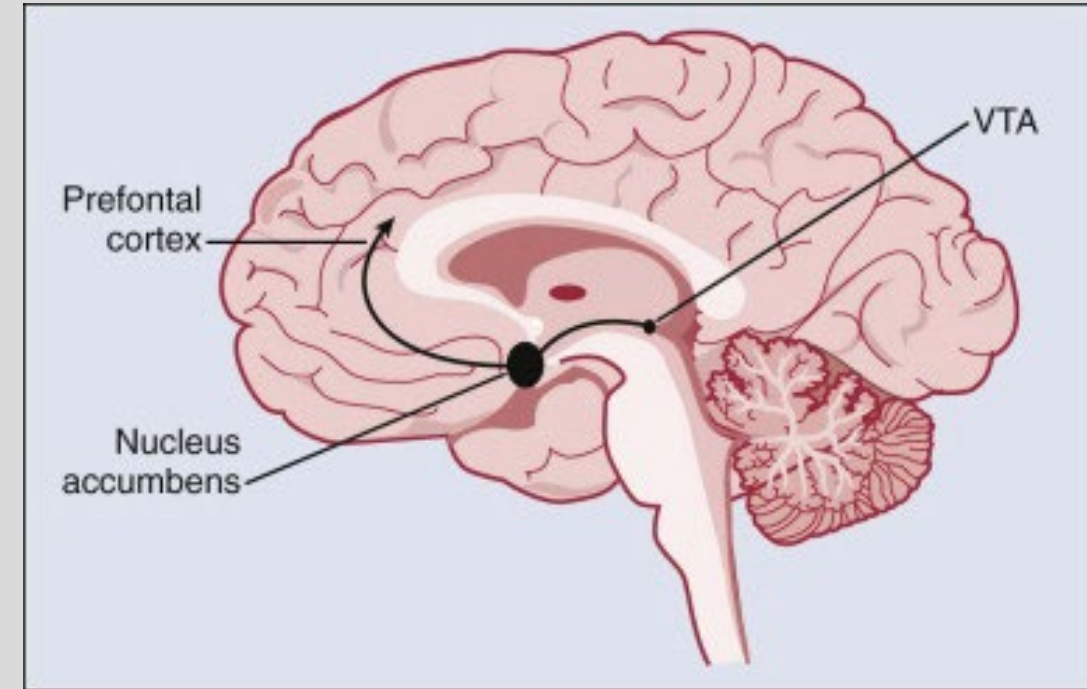


- Activation of Reward Pathway → Increase **Dopamine**
- “**Dopaminergic Reward Pathway**”
 - Dopamine is the messenger that travels from one area of the brain to another, activating pleasure and reward.
 - Repeated drug intake → repeated dopamine stimulation → permanent changes in the brain → compulsive drug intake

- Rang & Dale’s Pharmacology 8th edition

- Reward Pathway: Anatomy

- **“Mesolimbic Dopaminergic Pathway”**
 - Meso in Greek = middle (midbrain)
 - Mesolimbic = Midbrain + Limbic System
- **Midbrain**
 - **Ventral Tegmental Area (VTA): dopamine production**
- **Limbic Area**
 - Amygdala (emotions)
 - Hippocampus (memory)
 - **Nucleus Accumbens (NA)** in basal ganglia (habit formation, pleasure) . It is the place where gene expression alterations produced by excessive dopamine might underlie addiction.
 - Prefrontal Cortex (executive planning)

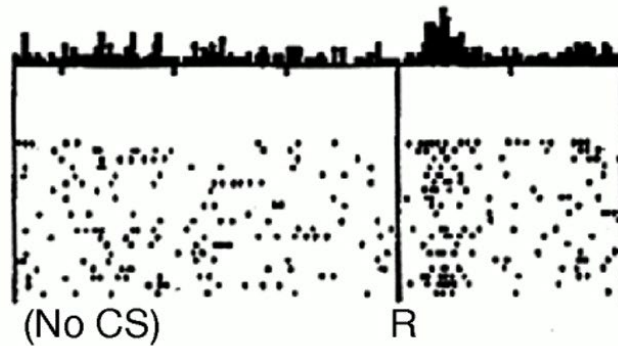


- Abelloff's Clinical Oncology, 5th edition

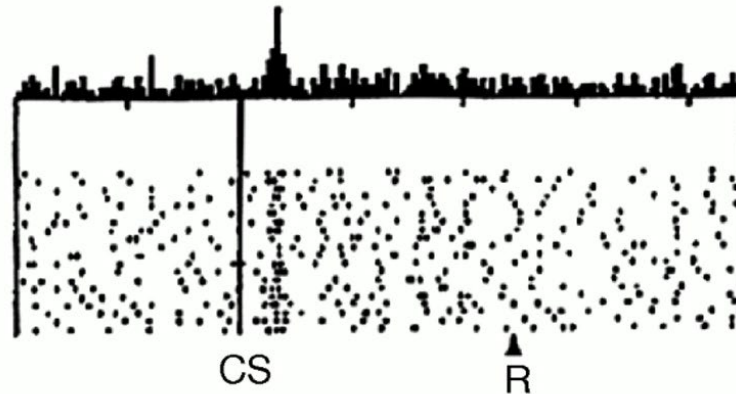
Dopaminergic neurons in VTA encode reward prediction error

Do dopamine neurons report an error
in the prediction of reward?

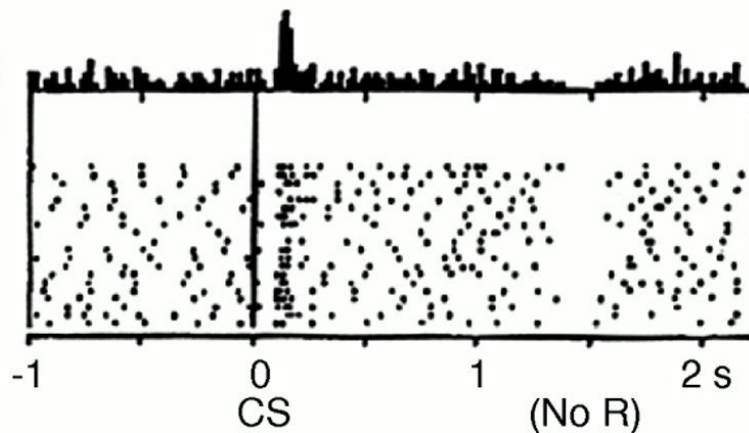
No prediction
Reward occurs



Reward predicted
Reward occurs



Reward predicted
No reward occurs



Recordings in monkey VTA

Initially VTA cells increase their firing rate as the monkey receives reward. The reward (R) happens at random times that the monkey cannot predict. Following this unpredictable reward, the VTA was activated.

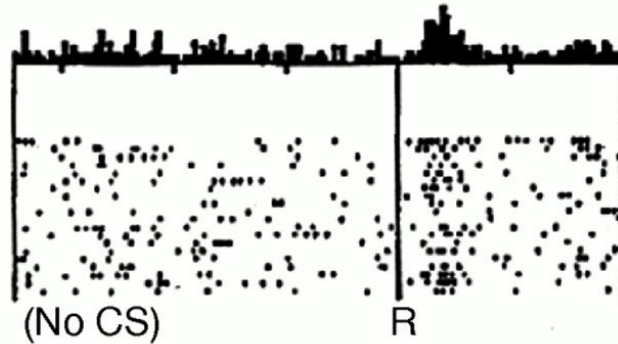
In the second set of experiments there was a light (Conditioned Stimulus or CS) that preceded the onset of the reward. This light predicts the reward. In this case, VTA neurons respond to the light and it stops responding to the reward. The idea is that as a stimulus becomes predictable, it stops activating the dopaminergic pathway.

In the third set of experiments the light went on but the reward was omitted. As the reward was omitted the VTA neurons suppressed their spontaneous activity.

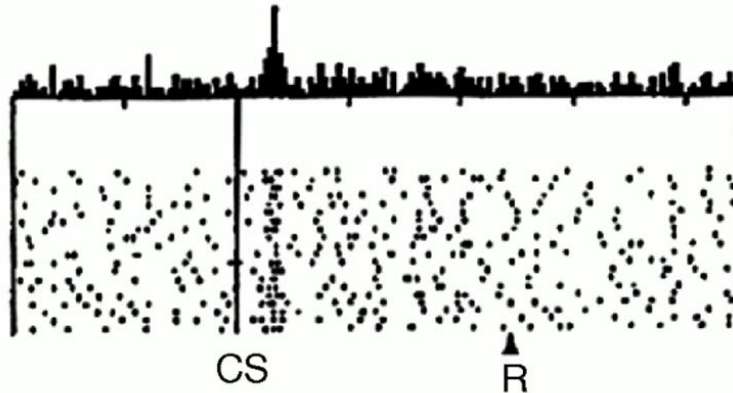
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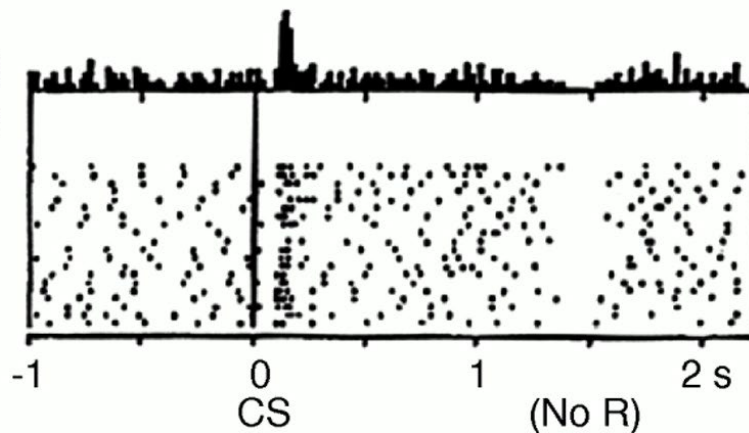
No prediction
Reward occurs



Reward predicted
Reward occurs



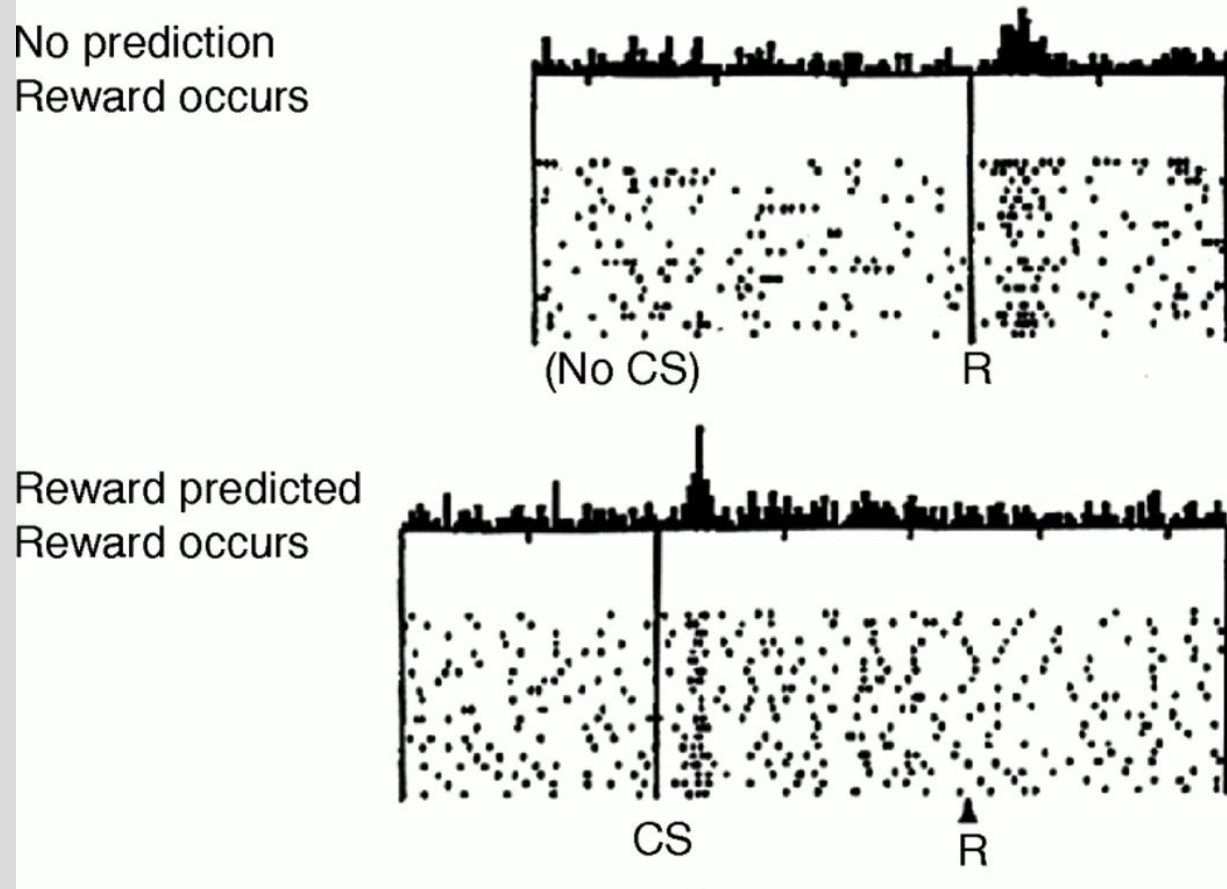
Reward predicted
No reward occurs



The dopaminergic system in the VTA represents the reward prediction error. If a stimulus is unpredictable, it would produce a large activation in the VTA. However, as a reward becomes more predictable, its activation of VTA is reduced.

Although activation of VTA might be pleasant, the purpose of the system is to learn how to predict high valuable stimuli. This framework of reward prediction error is used in artificial systems that learn to play chess among other applications of a technique called reinforcement learning.

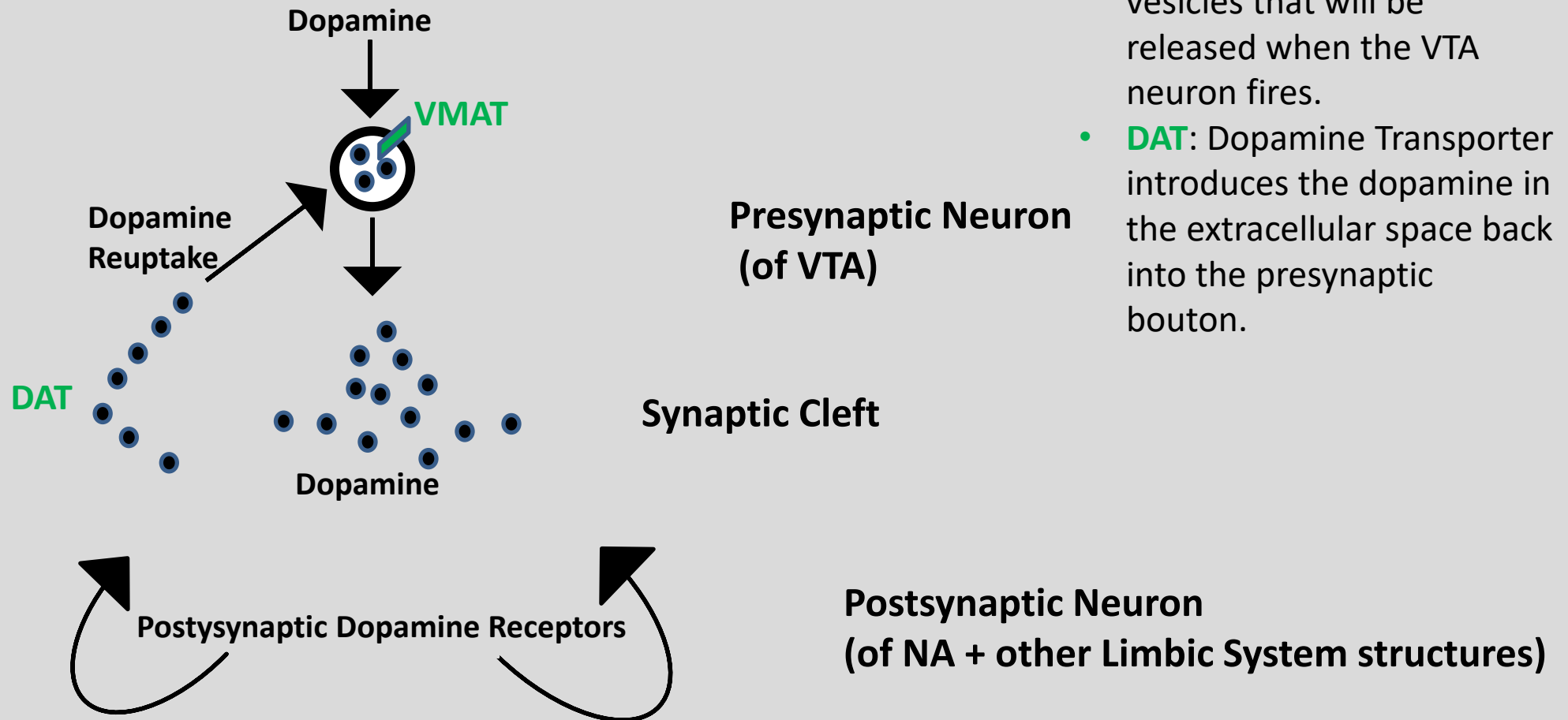
Drugs of abuse directly affect dopamine transmission



As a natural reward becomes more predictable, it stops the activation of the VTA and the dopaminergic release. However, drugs of abuse directly release dopamine from the axonal terminals of the VTA neurons or directly activate them and do not undergo the natural process of adaptation. They are a much stronger stimuli and there it lies its addiction potential.

A Neural Substrate of Prediction and Reward
Wolfram Schultz, Peter Dayan, P. Read Montague*, *Science* 1997

- Reward Pathway: Cellular Transmission



- Outline of Drugs

- **Stimulants**

- Amphetamines
- Cocaine
- Nicotine
- Caffeine

- **Depressants**

- Opioids
- Alcohol
- Benzodiazepines
- Barbiturates

- **Hallucinogens**

- Phencyclidine
- LSD
- Marijuana
- MDMA



- Stimulants: Overview

- **Stimulants:**
 - Speed up CNS activity
 - Mimic affects of endogenous agonists of **sympathetic nervous system**
 - Endogenous agonists = catecholamines (dopamine, norepinephrine, epinephrine)
 - Stimulants act like catecholamines → stimulants are **sympathomimetics**
 - Mechanism of sympathomimetic affect varies, depending on the drug
- **Stimulant Intoxication: General Symptoms**
 - Mood elevation, psychomotor agitation, insomnia, cardiac arrhythmias, tachycardia, anxiety
- **Stimulant Withdrawal: General Symptoms**
 - Post use-**crash**, depression, lethargy, increased appetite, sleep disturbances, vivid nightmares

- Stimulants: Amphetamines

- Amphetamines

- MOA:**

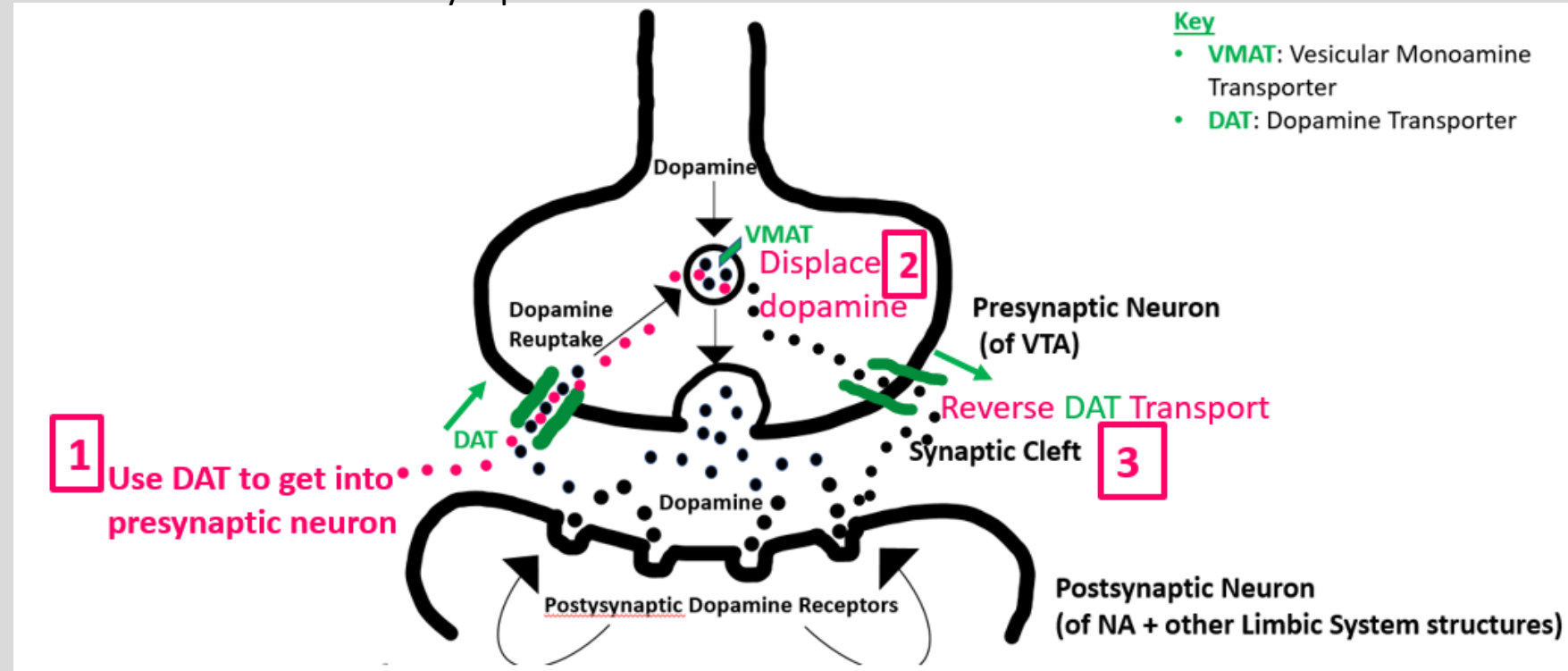
- Use **Dopamine Transporter** to get into presynaptic terminal
- In presynaptic neuron, utilize **VMAT** to enter neurosecretory vesicle, displacing dopamine from vesicle
- Dopamine concentration in presynaptic neuron reaches threshold, which **reverses action of DAT**
- Dopamine exits presynaptic neuron via reversed DAT into synaptic cleft

- Intoxication:**

- See General Symptoms
- Mydriasis (pupillary dilation)
- Severe: cardiac arrest, seizures

- Withdrawal:**

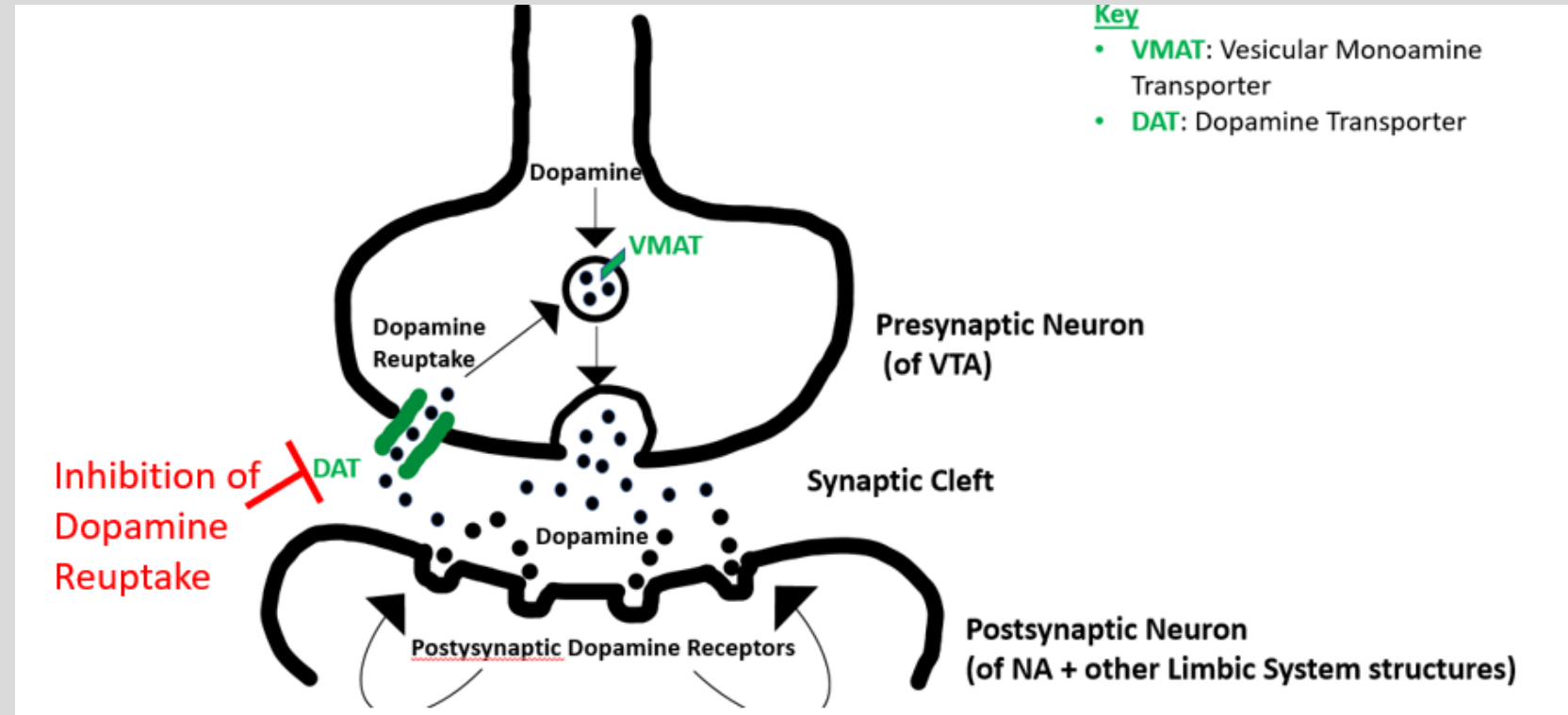
- See General Symptoms



- Stimulants: Cocaine

Cocaine

- Inhibits the reuptake of dopamine by inhibiting DAT.
- The inhibition of the dopamine reuptake will produce large activation of the postsynaptic receptors.
- **Coronary artery vasospasm**
Mydriasis (pupillary dilation)
- **Hallucinations (tactile)**
- Paranoia
- **Nasal septum perforation**
produced by tissue death as a consequence of vasospasm.
- **Withdrawal:**
 - See General Symptoms

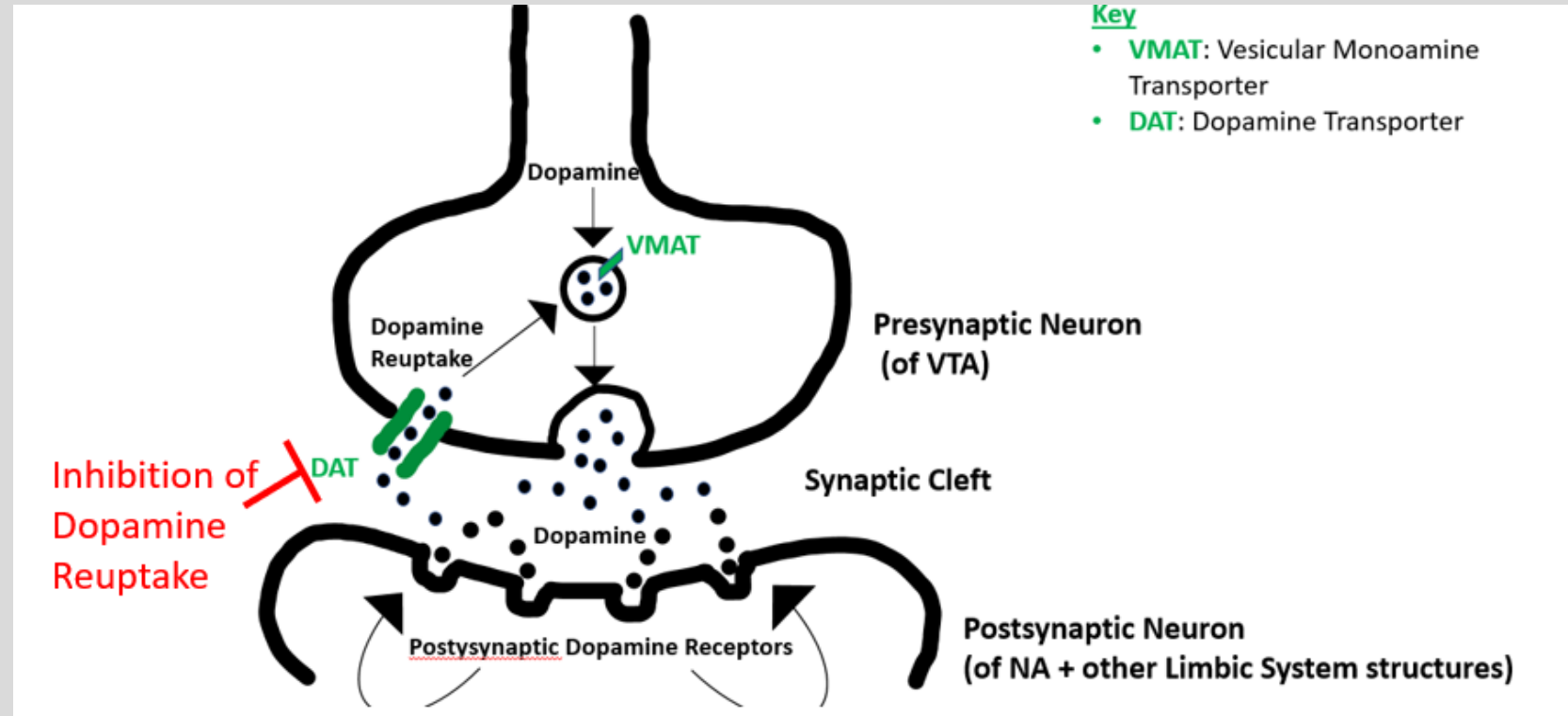


- Stimulants: Cocaine

Cocaine

- **MOA:**

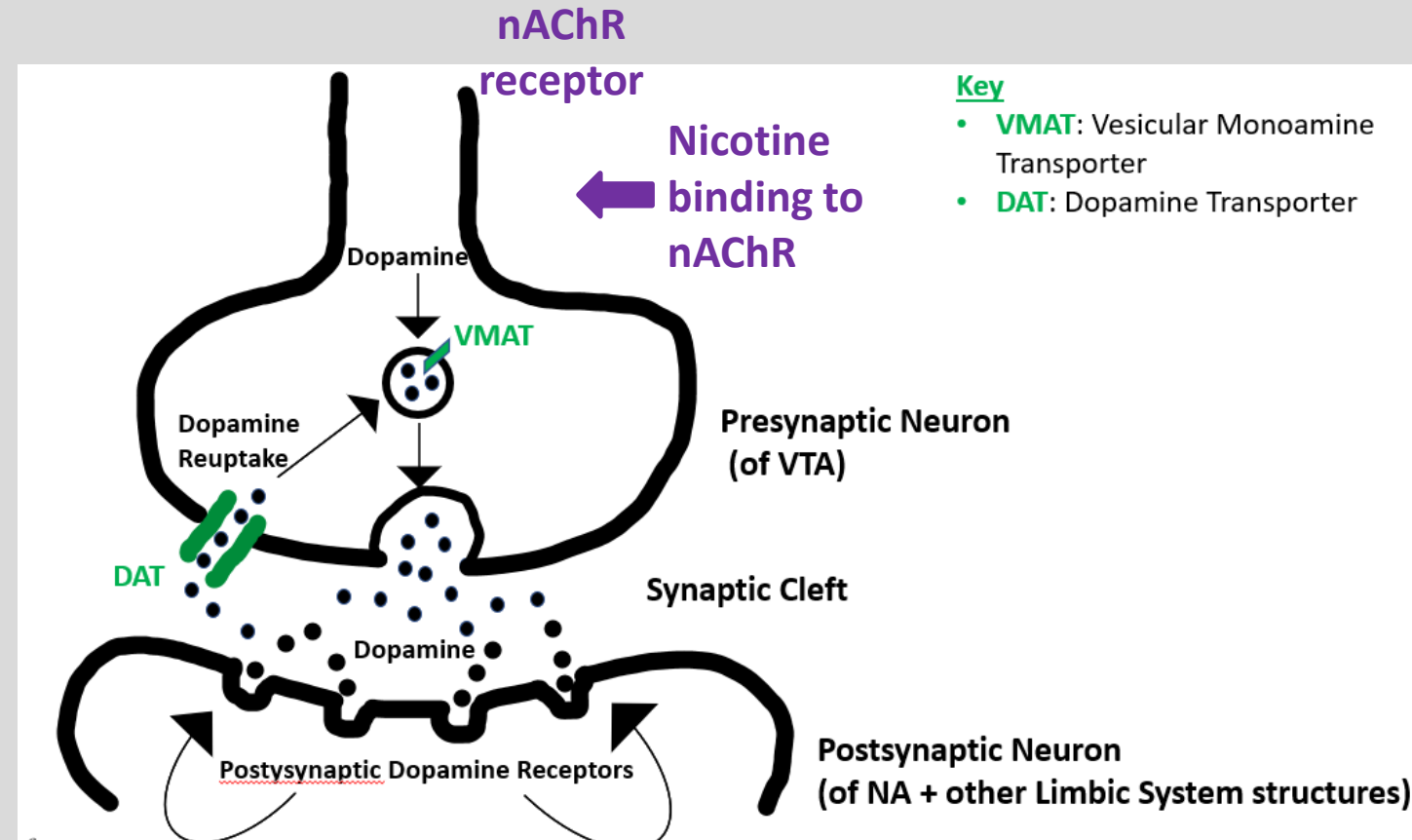
- Inhibits the reuptake of dopamine and **other catecholamines, epinephrine (adrenaline), and norepinephrine**
- Excess of catecholamines: **Coronary artery vasospasm**
→ Sudden cardiac death
- Treatment: alpha blocker or benzo; **do NOT give beta blocker**
- If the beta receptors are blocked by the beta blocker, the excess catecholamines will bind to alpha receptors causing further **vasoconstriction**



- Stimulant: Nicotine

Nicotine

- **MOA:**
 - **Binds nicotinic ACh receptors** in Ventral Tegmental Area (VTA).
 - Activation of the nicotinic receptor will depolarize the dopaminergic axonal bouton.
 - The depolarization of the bouton **increases dopamine** release.
- **Intoxication:**
 - Restlessness
- **Withdrawal:**
 - Irritability, anxiety, difficulty concentrating



- Goldman-Cecil Medicine, 25th edition

- Depressant: Overview

- **Depressants:**
 - “Downers” that lower arousal levels
 - Depressant effects largely mediated by increasing inhibitory neurotransmission (GABA) and decreasing excitatory neurotransmission (glutamate)
- **Depressant Intoxication: General Symptoms**
 - Mood elevation, decreased anxiety, sedation, behavioral disinhibition, respiratory depression
- **Depressant Withdrawal: General Symptoms**
 - Anxiety, tremor, seizures, insomnia

- Depressant: Opioids

Opioids

- MOA:

- VTA contains GABA-ergic interneurons that inhibit the dopaminergic neurons.
- **Opioids bind to mu-opioid receptor in the GABAergic interneurons and inhibit those cells.**
- There is a reduction of GABA inhibition of the dopaminergic neurons
- Reduction of inhibitions produces **increased dopamine** release

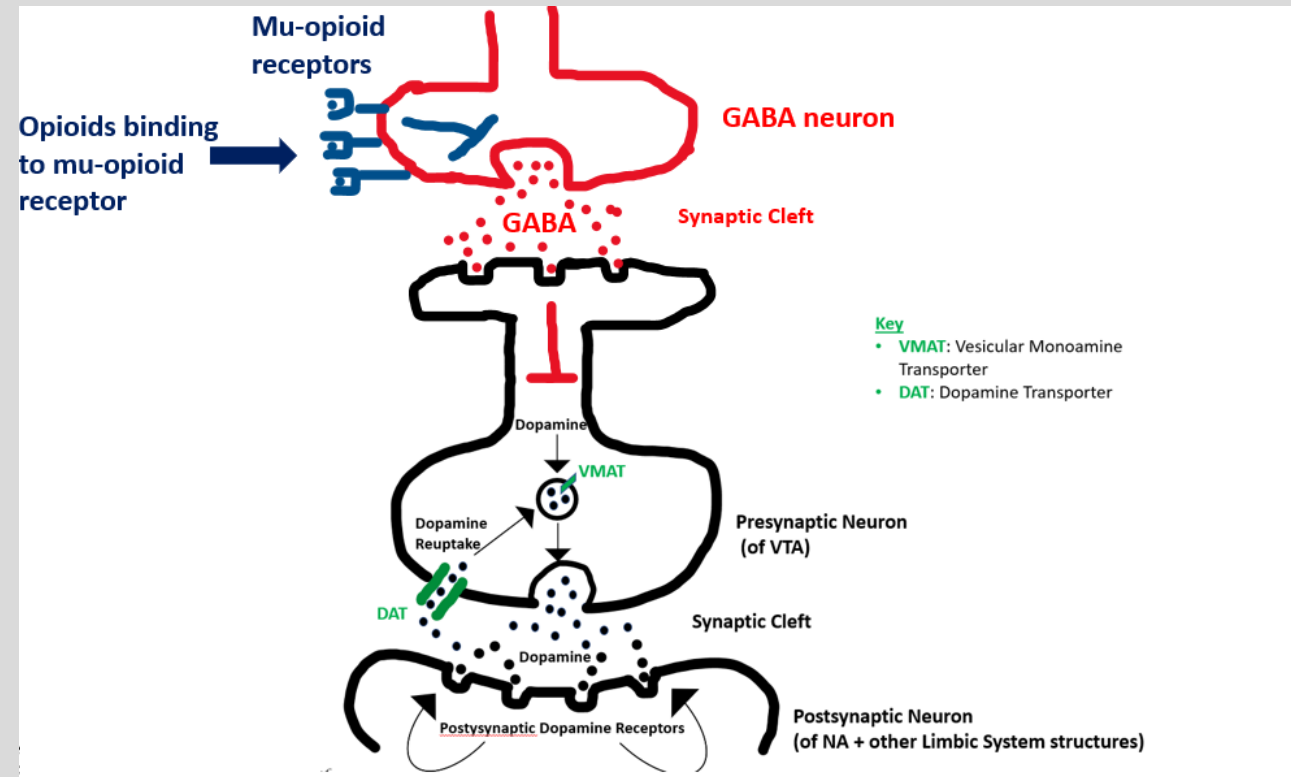
- **Intoxication:**

- Respiratory & CNS depression
- Miosis, **constipation**, itching

- **Withdrawal:**

- **Diarrhea**, sweating, dilated pupils, piloerection (cold turkey), fever, rhinorrhea, **yawning**, nausea, stomach cramps (flu-like symptoms).
- Lacrimation is very characteristic of opioid withdrawal.

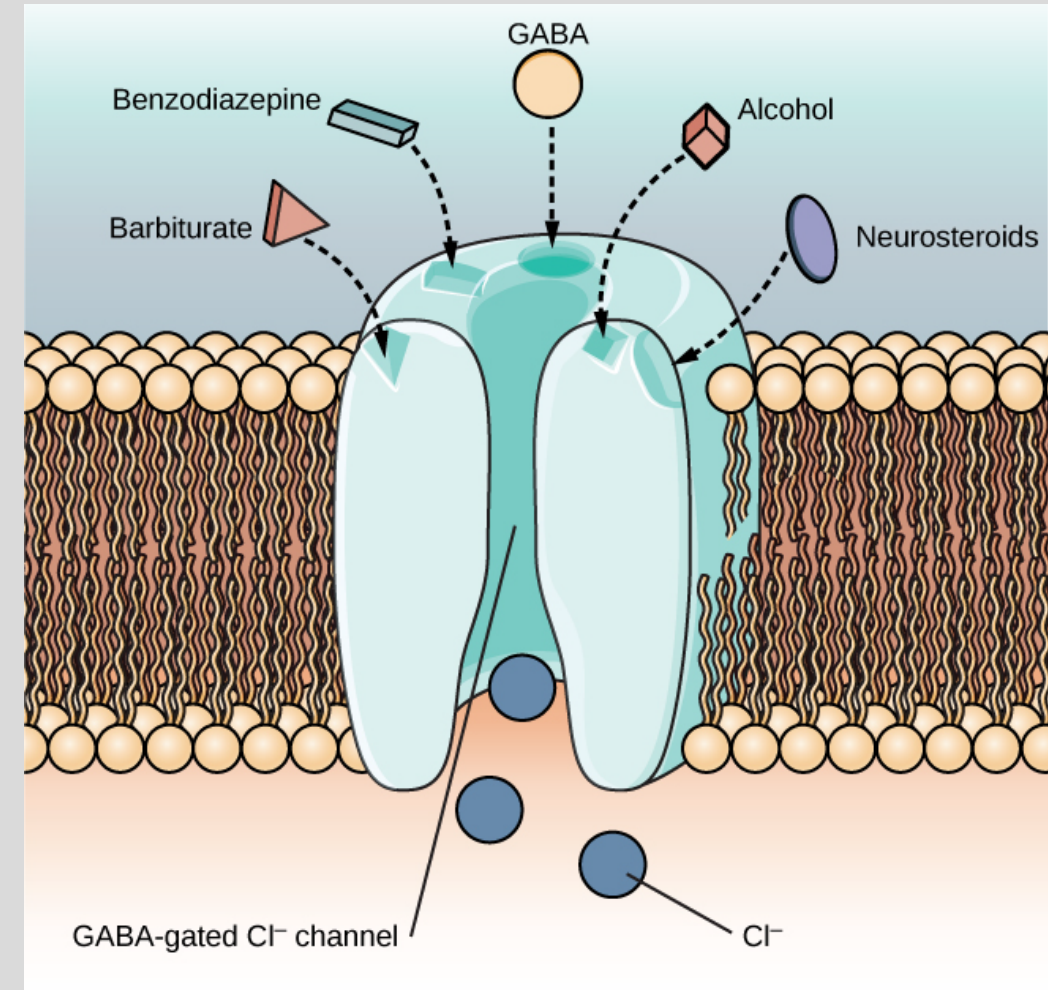
- Meyler's Side Effects of Drugs, 16th edition



- An infant born to a mother with a substance use disorder is at risk for withdrawal
- Naloxone is a non-selective and competitive opioid receptor antagonist, used for respiratory depression

- Preview: Alcohol, Benzos, and Barbs

- Alcohol, Benzodiazepines, and Barbiturates are all depressants that elicit their “depressant” effects by binding to allosteric sites on **GABA receptor**.
- GABA receptor is a **chloride gated channel**
 - Binding to GABA receptor → Cl⁻ channel opens → Cl⁻ influx → hyperpolarization of membrane → **inhibitory affect**

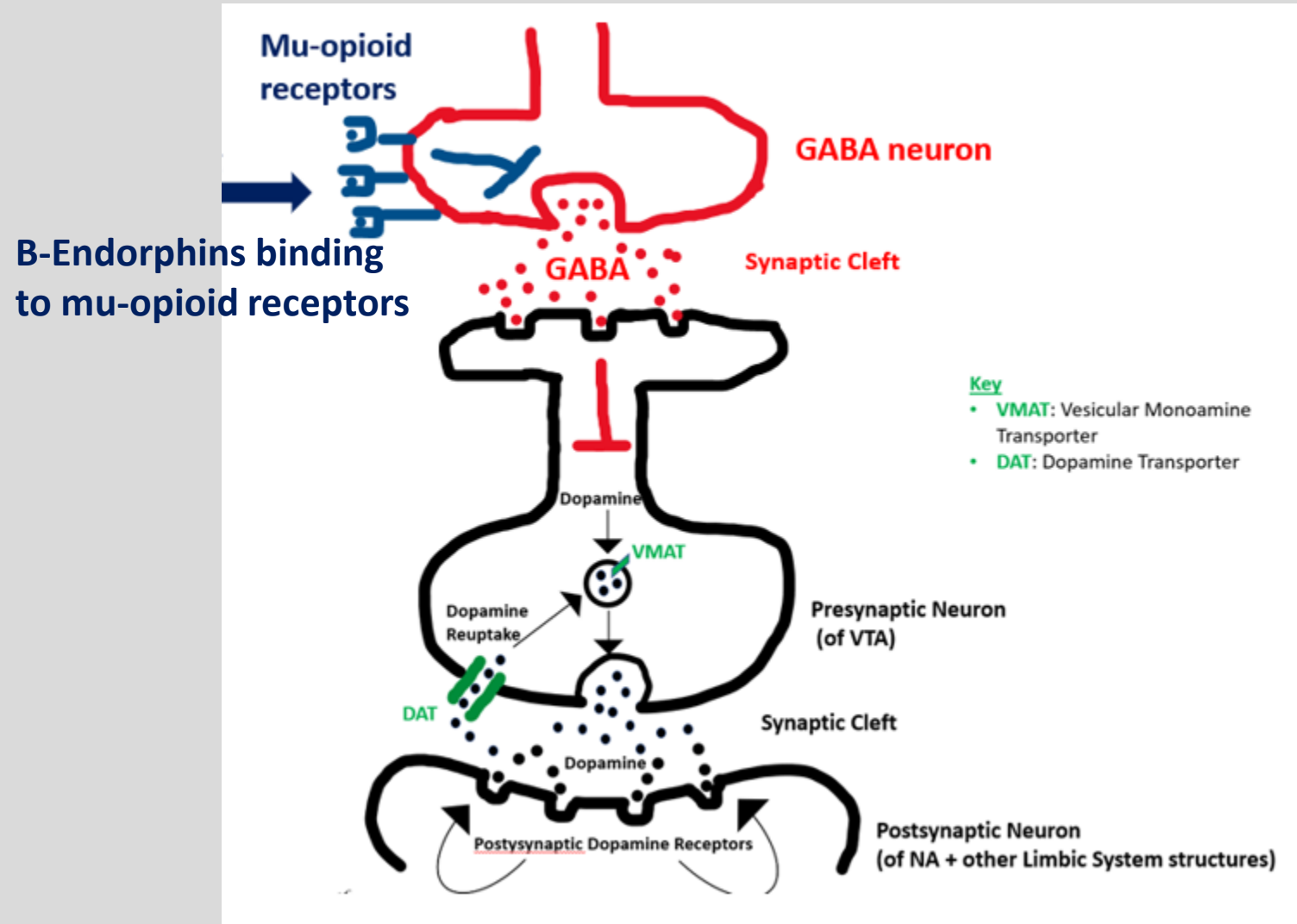


- Depressant: Alcohol

Alcohol

• MOA:

- Increases release of **B-endorphins**
- B-endorphins bind to mu-opioid receptors → inhibits GABA → disinhibits dopamine → **dopamine released**



- Massachusetts General Hospital Comprehensive Clinical Psychiatry, 2nd edition

- Depressant: Alcohol

Alcohol

- **Intoxication:**
 - (ABCDES: ataxia, blackouts, coma, disinhibition, emotional lability, slurred speech)
- **Long Term Abuse Complications:**
 - Alcoholic cirrhosis, hepatitis, pancreatitis, peripheral neuropathy, testicular atrophy
- **Mild Withdrawal:**
 - See general nonspecific
- **Severe Withdrawal:**
 - Autonomic hyperactivity (tachycardia, tremors, anxiety, seizures) & DT
 - **Delirium Tremens:** life-threatening 2-4 day peak after last drink; common in hospital setting (2-4 days post op)
 - Treat DT: benzo- chlordiazepoxide, lorazepam, diazepam
 - **Alcoholic hallucinosis:** distinct condition characterized by **visual hallucinations** 12-48 hrs after last drink

- Emergency Medicine, 2nd edition

- Depressant: Benzodiazepines

Benzodiazepines

- **MOA:**

- Binds an allosteric site on GABA A receptor
 - → increases **frequency** of Cl⁻ channel opening → hyperpolarization → inhibitory/depressant affect
 - “frenzodiazepines increase frequency”
 - → **inhibits GABA interneuron** from releasing GABA → disinhibition dopamine → **increased dopamine release**

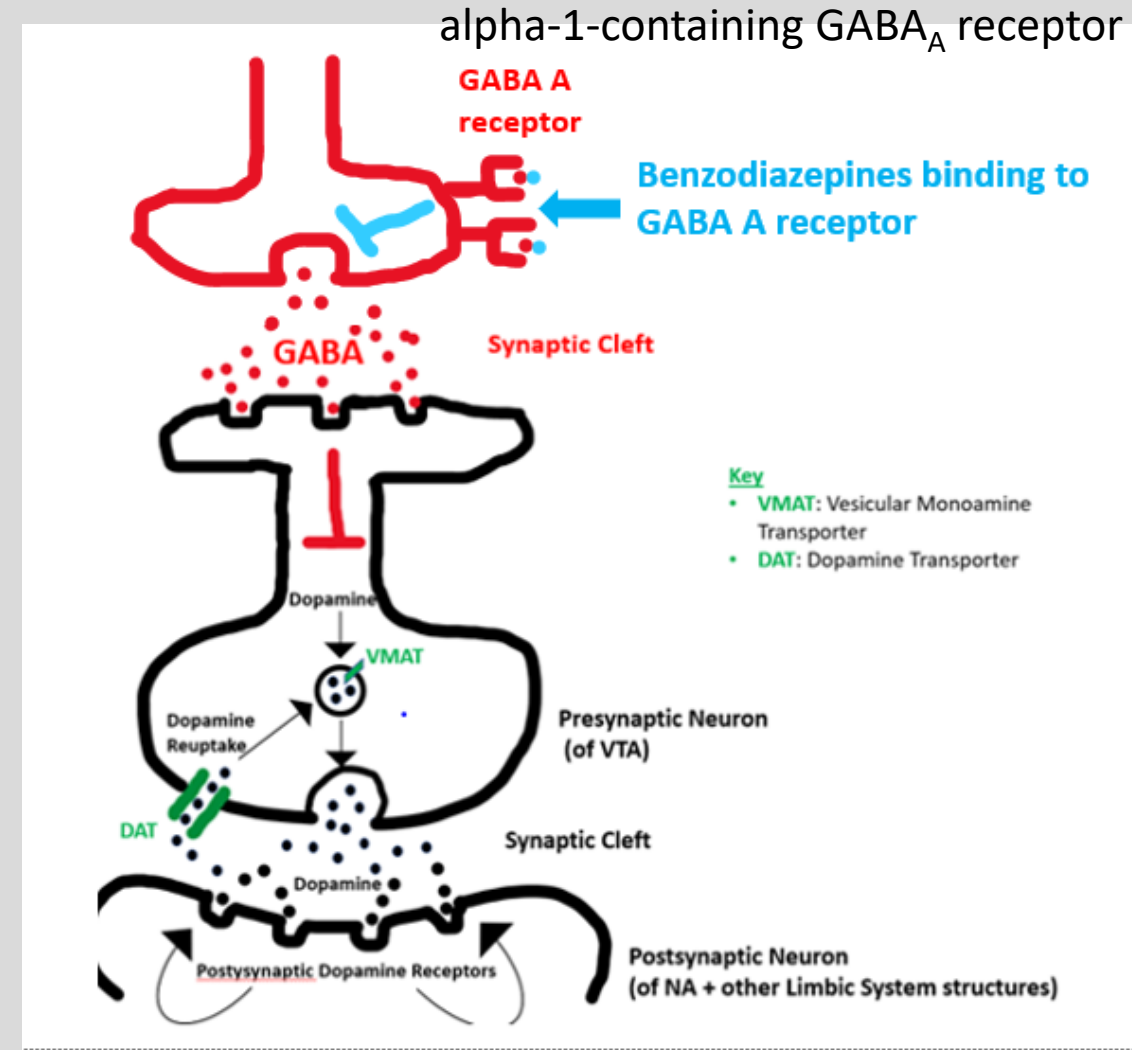
- **Intoxication:**

- Minor resp. depression, **additive CNS affects with alcohol**
 - **Treatment:** flumazenil: competitive antagonist at GABA benzo receptor

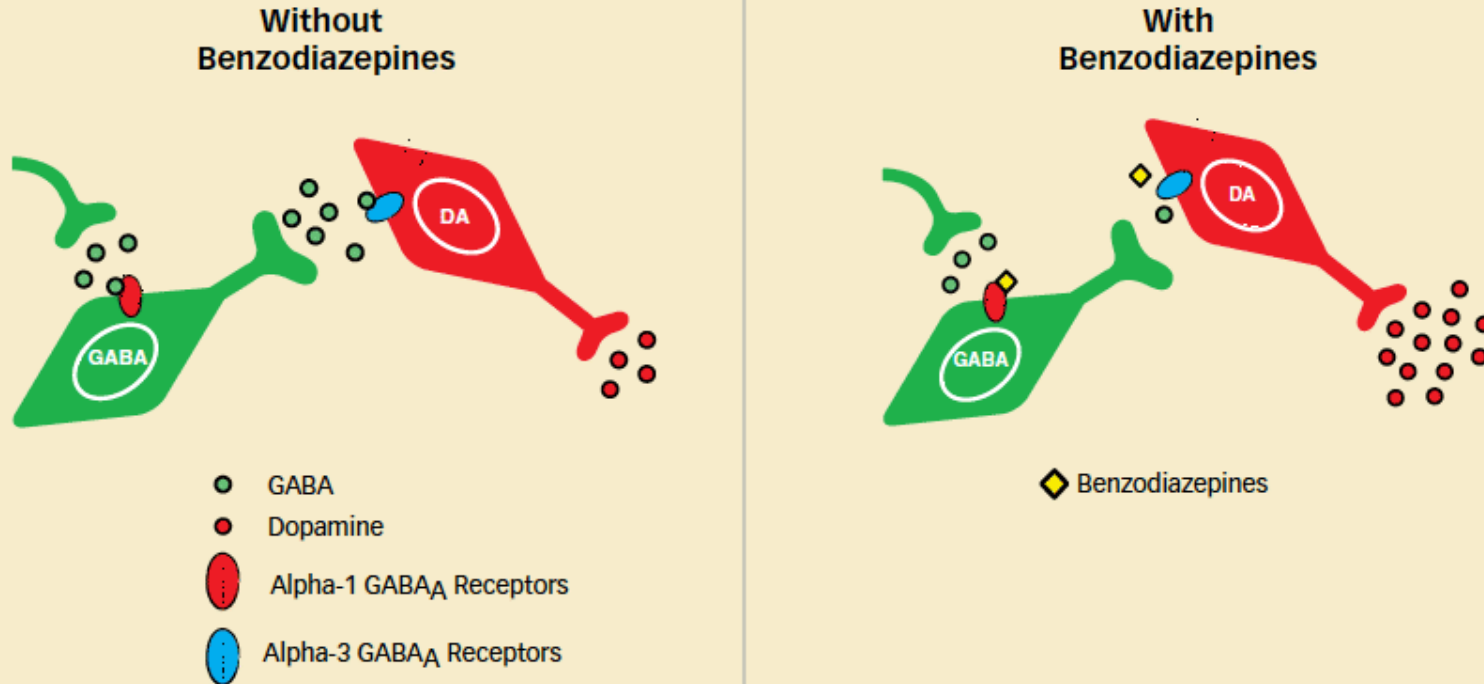
- **Withdrawal:**

- Sleep disturbance, depression, rebound anxiety, seizure

- Meyler’s Side Effects of Drugs, 16th edition



- Differences in GABA receptor subunits underlie addictive properties of benzodiazepines



- Both dopaminergic neurons and GABAergic neurons in the VTA have GABA_A receptors.
- However they express different subunits. The dopaminergic neurons express the Alpha-3 GABA_A subunit.
- The GABAergic neurons express the Alpha-1 GABA_A subunit.
- Benzodiazepines bind to the Alpha-1 subunit.
- Therefore Benzodiazepines will selectively inhibit the GABA interneurons, producing activation of the DA neurons.

<https://archives.drugabuse.gov/news-events/nida-notes/2012/04/well-known-mechanism-underlies-benzodiazepines-addictive-properties>

- Depressant: Barbiturates

Barbiturates

- **MOA:**

- Binds an allosteric site on GABA A receptor
 - → **increases duration** of Cl channel opening
 - “*barbiturates increase duration*”
 - → presumably similar reward mechanism as seen in benzos, though not fully understood

- **Intoxication:**

- Low safety margin, **marked respiratory depression, CV depression, CNS depression** (exacerbated by alcohol use)
- **Treatment:** symptom management (assist respiration, increase BP)

- **Withdrawal:**

- Delirium, life-threatening CV collapse

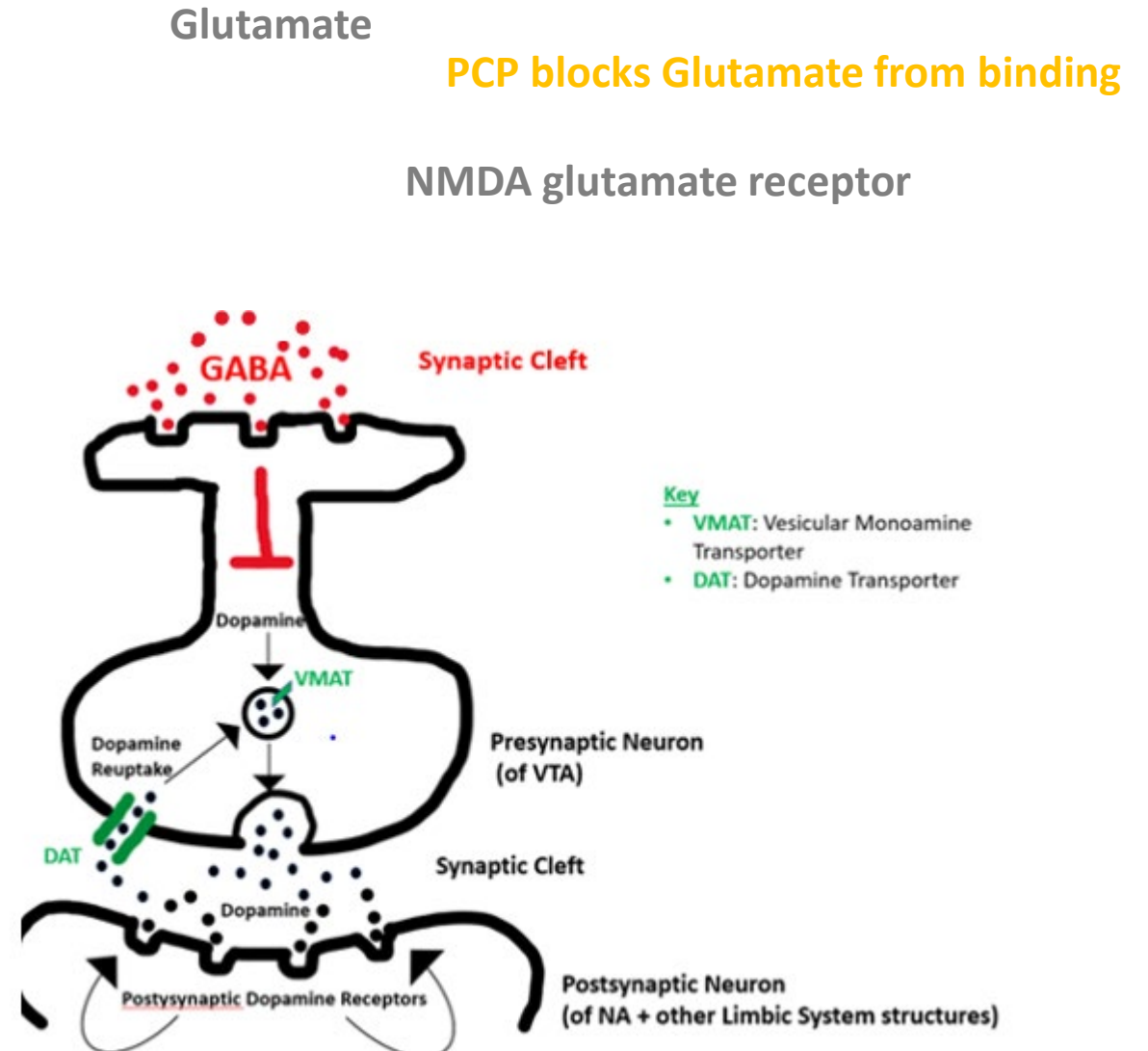
- Hallucinogen: Phencyclidine (PCP)

Phencyclidine (PCP)

“Angel Dust”

- **MOA:**
 - **NMDA glutamate receptor antagonist**
→ inhibits GABA → disinhibits dopamine
- **Intoxication:**
 - **Violence** (homicidal tendencies), impulsivity, psychomotor agitation, **nystagmus**, tachycardia, HTN, analgesia, psychosis, delirium, seizures; most common complication: **trauma**
- **Withdrawal:**
 - Not well studied

- Rang & Dale's Pharmacology 8th edition



- Hallucinogen: LSD

LSD (Lysergic Acid Diethylamide)

- **MOA:**
 - Serotonin 5HT₂ partial agonist
- **Intoxication:**
 - Sensory misperception
 - Panic attacks, anxiety
 - Sympathomimetic affects
- Rang & Dale's Pharmacology 8th edition



- Hallucinogen: Marijuana (Cannabinoid)

Marijuana

• MOA:

- Active ingredient: **tetrahydrocannabinol (THC)**
- Binds to and activates **CB1 receptor** on GABA interneuron → inhibits GABA → disinhibits dopamine

• Intoxication:

- Distorted perception of time, **delayed reaction time**, impaired judgement, increased appetite, dry mouth, hallucinations

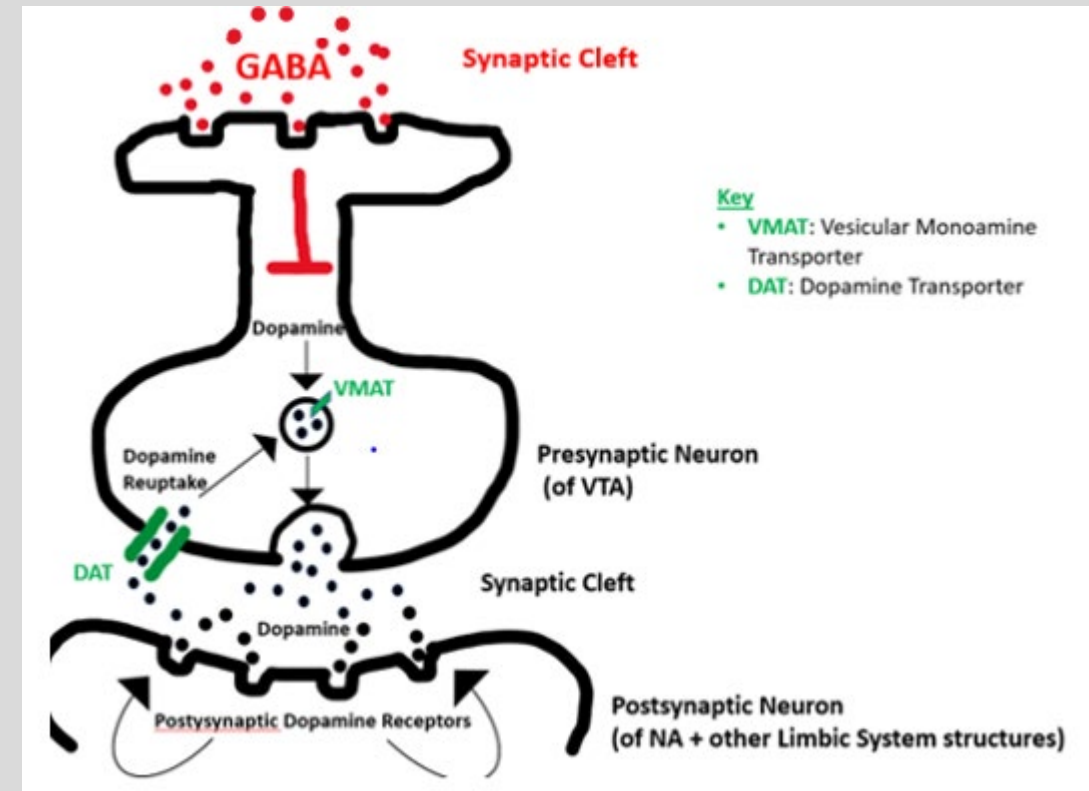
• Withdrawal:

- Irritability, anxiety, depression, insomnia, restlessness, decreased appetite
- After discontinuation: detected in **urine 7-10 days** in casual user; **1 month** in chronic user

- Goldman-Cecil Medicine, 25th edition

Marijuana binds
to CB1 receptor

endocannabinoid
receptor (CB1)



- Hallucinogen: MDMA (Ecstasy)

MDMA (Ecstasy)

- MOA:

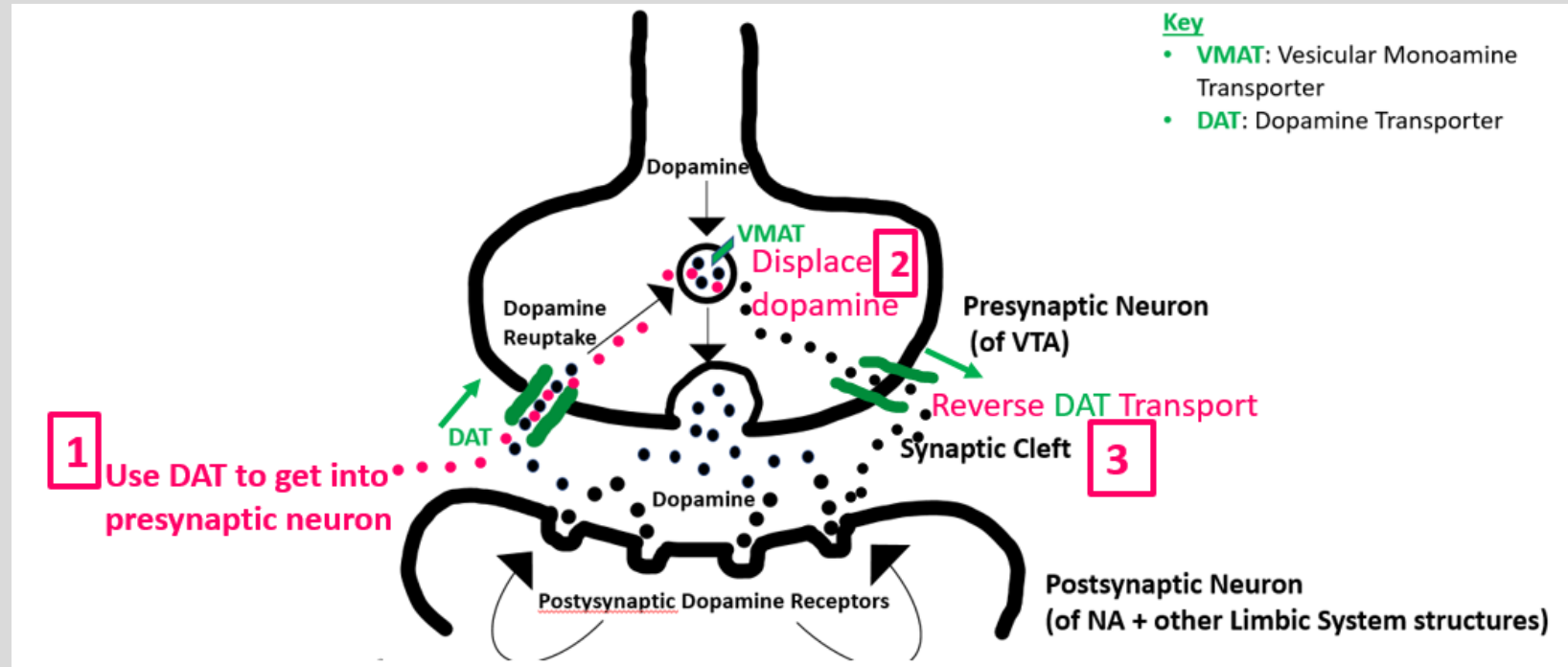
- Amphetamine derivative → very similar to that of amphetamines
- Works on dopamine system but mainly on **serotonin** neurotransmitter
 - Uses **SERT** (serotonin transporter) to get into presynaptic neuron → displaces serotonin from VMAT → increased concentration serotonin reverses SERT transporter

- Intoxication:

- Life threatening: HTN, tachycardia, **hyperthermia**, hyponatremia, **serotonin syndrome**
- Treat: normal saline & cooling
- Metabolites of MDMA interfere with its breakdown so a second dose will have a stronger effect.

- Withdrawal:

- Depression, fatigue, change in appetite, difficulty concentrating, anxiety



- Swaiman's Pediatric Neurology, 6th edition
- <https://www.drugabuse.gov/publications/research-reports/mdma-ecstasy-abuse/what-are-effects-mdma>

- Conclusions

- The dopaminergic mesolimbic pathway is the neurobiological basis for drug addiction, as it pertains to reinforcing mechanisms
- The three broad classes of drugs of abuse include stimulants, depressants, and hallucinogens, each of which variably target the dopaminergic mesolimbic pathway
- Detailed review of symptomatology, (intoxication/withdrawal clues) guides us in differentiating between drugs of abuse in a clinical setting
- Epidemiological evidence suggests that drug addiction is a serious and growing national problem, making it all the more essential to understand the targeted neuropharmacological pathways

- Case Break

Case #1

A 30 yo male presents to the ED with chest pain that radiates down his left arm. Past medical history is unremarkable. Physical exam reveals a hole in the cartilaginous membrane that separates his nostrils. Initial lab workup, including EKG and cardiac enzymes, confirms a STEMI. Further diagnostic testing in the cath lab shows occlusions in 3 coronary vessels. Which MOA explains the reinforcing properties of the drug you suspect he is abusing?

- A) The binding to mu-opioid receptors, which inhibits GABA interneuron, thereby disinhibiting dopamine
- B) The binding to CB1 receptors, which inhibits GABA, thereby disinhibiting dopamine
- C) Inhibition of the dopamine reuptake transporter, thereby increasing dopamine in synaptic cleft
- D) Using dopamine reuptake transporter to enter presynaptic neuron, displacing dopamine from vesicles, thereby increasing dopamine in synaptic cleft

- Case Break: Answer

Case #1: Answer

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- Case Break

Case #2

A 55 yo man is found on the subway platform after being badly beaten. He is brought in by EMS to the ED, where he is AAO x3, though frustrated that he lost the fight on the platform. Vitals are stable, laboratory workup is pending, and imaging studies reveal a fractured nasion and R zygomatic arch. He is admitted for further observation. Three days later, he is notably confused. Vitals show that he is febrile and tachycardic. Physical exam shows a hand tremor and tongue fasciculations. You suspect this patient is:

- a) An alcoholic
- b) A meth addict
- c) A heroin user
- d) Meningitic

- Case Break Answer

Case #2 Answer

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- Case Break

Case #3

Dopamine production, packaging, and subsequent release is accomplished in this brain structure: _____, which is a target of many addictive drugs.

- a) Prefrontal Cortex
- b) Synaptic Cleft
- c) Nucleus Accumbens
- d) Ventral Tegmental Area

- Case Break Answer

Case #3: Answer

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Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>